

Unit 7

Partial Differential Equations of the First Order

Definition

An equation involving one or more partial derivatives of an unknown function of two or more independent variables is known as a partial differential equation.

Examples

$$1. \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = z + xy$$

$$2. \left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial^3 z}{\partial y^3} = 2x \left(\frac{\partial z}{\partial x}\right)$$

$$3. z \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = x$$

$$4. \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = xyz$$

$$5. \frac{\partial^2 z}{\partial x^2} = \left(1 + \frac{\partial z}{\partial y}\right)^{\frac{1}{2}}$$

$$6. y \left\{ \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right\} = z \frac{\partial z}{\partial y}$$

Order of Partial Differential Equation (PDE)

The order of the highest order derivative occurring in a partial differential equation (PDE) is called the order of the PDE.

Example

The equation (1), (3), (4) and (6) are first order pde, (5) is of second order and (2) is of third order pde in the above examples.

Degree of a PDE

The degree of a PDE is the degree of the highest order derivative present in the pde after the equation has been rationalized, that is, made free from radicals and fractions.

Examples

In the above examples, equations (1), (2), (3) and (4) are of first degree and equations (5) and (6) are of second degree.

Linear and Non-linear pde

A pde is said to be linear if the dependent variable and its partial derivatives occur only in the first degree and are not multiplied. A pde which is not linear is called a non-linear pde.

Examples

In above examples, equations (1) and (4) are linear while (2), (3), (5) and (6) are non-linear pdes.

Notations

In case of two independent variables, we assume them to be x and y and z is assumed to be dependent variable. In a partial differential equation with two independent variables we use the following notations.

$$\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial^2 z}{\partial x^2} = r, \frac{\partial^2 z}{\partial x \partial y} = s, \frac{\partial^2 z}{\partial y^2} = t.$$

Also,

$$u_x = \frac{\partial u}{\partial x}, u_y = \frac{\partial u}{\partial y}, u_{xx} = \frac{\partial^2 u}{\partial x^2}, u_{xy} = \frac{\partial^2 u}{\partial x \partial y}, u_{yy} = \frac{\partial^2 u}{\partial y^2}$$

are used as notation for partial differential equations.

Origin/Formation of PDE

Rule I [By Eliminating Arbitrary Constants]

Consider the Equation

$$F(x, y, z, a, b) = 0 \quad \dots \quad (1)$$

where a and b are arbitrary constants. Let z be a function of two independent variables x and y .

Differentiating (1) partially w.r.t x and y , respectively, we get

$$\frac{\partial F}{\partial x} + p \frac{\partial F}{\partial z} = 0 \text{ and } \frac{\partial F}{\partial y} + q \frac{\partial F}{\partial z} = 0 \quad \dots \quad (2)$$

Now, eliminating two arbitrary constants a and b from (1) and (2), we may get an equation of the form

$$f(x, y, z, p, q) = 0 \quad \dots \quad (3),$$

which is a partial differential equation of the first order.

Three Different Situations

Situation I

When the number of arbitrary constants is less than the number of independent variables, then the elimination of arbitrary constants generally gives more than one PDE of first order.

Example Form a PDE from $z = ax + y$

Solution

The given function is

$$z = ax + y \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t. x and y , we get

$$\frac{\partial z}{\partial x} = a \quad \dots \quad (2)$$

$$\text{and } \frac{\partial z}{\partial y} = 1 \quad \dots \quad (3)$$

Eliminating a between (1) and (2) we get

$$z = x \frac{\partial z}{\partial x} + y \quad \dots \quad (4)$$

Since (3) contains no arbitrary constants, it is also a differential equation.

Thus, we get (3) and (4) as two partial differential equations of the first order.

Situation II

When the number of arbitrary constants is equal to the number of independent variables then the elimination of arbitrary constants gives a unique PDE of first order.

Example Eliminate a and b from $az + b = a^2x + y$ to form a PDE.

Solution

The given function is

$$az + b = a^2x + y \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t. x and y , we get

$$a \frac{\partial z}{\partial x} = a^2 \quad \dots \quad (2)$$

$$\text{and } a \frac{\partial z}{\partial y} = 1 \quad \dots \quad (3)$$

Eliminating a from (2) and (3), we get

$$\frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} = 1,$$

which is a unique PDE of first order.

Situation III

When the number of arbitrary constants is greater than the number of independent variables, then the elimination of arbitrary constants produces a PDE of order greater than one.

Example Form a PDE by eliminating a, b, c from

$$z = ax + by + cxy.$$

Solution

The given function is

$$z = ax + by + cxy = 0 \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t x and y , we get

$$\frac{\partial z}{\partial x} = a + cy \quad \dots \quad (2)$$

$$\frac{\partial z}{\partial y} = b + cx \quad \dots \quad (3)$$

Again, from (2) and (3)

$$\frac{\partial^2 z}{\partial x^2} = 0, \frac{\partial^2 z}{\partial y^2} = 0 \quad \dots \quad (4)$$

$$\text{and } \frac{\partial^2 z}{\partial x \partial y} = c \quad \dots \quad (5)$$

Again, (2) implies

$$x \frac{\partial z}{\partial x} = ax + cxy$$

and (3) implies

$$y \frac{\partial z}{\partial y} = by + cxy$$

Adding these two we get

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = ax + by + cxy + cxy$$

using (1) and (2), we get

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z + xy \frac{\partial^2 z}{\partial x \partial y} \quad \dots \quad (6)$$

Thus, we get three PDEs given by (4) and (6) which are all of order two.

Examples

1. Form PDE by eliminating arbitrary constants a and b from

$$z = ax + by + ab.$$

Solution

The given function is

$$z = ax + by + ab \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t. x and y we get

$$\frac{\partial z}{\partial x} = a \quad \dots \quad (2)$$

$$\text{and } \frac{\partial z}{\partial y} = b \quad \dots \quad (3)$$

Using (2) and (3) in (1), we get

$$z = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y}$$

which the required PDE of first order.

2. Form a PDE by eliminating h and k from

$$(x - h)^2 + (y - k)^2 + z^2 = \lambda^2$$

solution

The given function is

$$(x - h)^2 + (y - k)^2 + z^2 = \lambda^2 \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t. x and y , we get

$$\begin{aligned} 2(x - h) + 2z \frac{\partial z}{\partial x} &= 0 \\ \text{or, } x - h &= -z \frac{\partial z}{\partial x} \quad \dots \quad (2) \end{aligned}$$

and

$$\begin{aligned} 2(y - k) + 2z \frac{\partial z}{\partial y} &= 0 \\ y - k &= -z \frac{\partial z}{\partial y} \quad \dots \quad (3) \end{aligned}$$

Using (2) and (3) in (1), we get

$$\begin{aligned} z^2 \left(\frac{\partial z}{\partial x} \right)^2 + z^2 \left(\frac{\partial z}{\partial y} \right)^2 + z^2 &= \lambda^2 \\ \text{or, } z^2 \left[\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1 \right] &= \lambda^2 \end{aligned}$$

which is the required PDE of first order.

3. Find a pde by eliminating a, b and c from

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Solution

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t x and y , we get

$$\begin{aligned} \frac{2x}{a^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial x} &= 0 \\ c^2 x + a^2 z \cdot \frac{\partial z}{\partial x} &= 0 \quad \dots \quad (2) \end{aligned}$$

And

$$\begin{aligned} \frac{2y}{b^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial y} &= 0 \\ \text{or, } c^2 y + b^2 z \cdot \frac{\partial z}{\partial y} &= 0 \quad \dots \quad (3) \end{aligned}$$

Again, differentiating (2) w.r.t. x and (3) w.r.t. y , we get

$$c^2 + a^2 \left(\frac{\partial z}{\partial x} \right)^2 + a^2 z \frac{\partial^2 z}{\partial x^2} = 0 \quad \dots \quad (4)$$

And

$$c^2 + b^2 \left(\frac{\partial z}{\partial y} \right)^2 + b^2 z \frac{\partial^2 z}{\partial y^2} = 0 \quad \dots \quad (5)$$

From (2)

$$c^2 = -\frac{a^2 z}{x} \cdot \frac{\partial z}{\partial x} \quad \dots \quad (6)$$

Using this in (4), we get

$$\begin{aligned} & -\frac{a^2 z}{x} \cdot \frac{\partial z}{\partial x} + a^2 \left(\frac{\partial z}{\partial x} \right)^2 + a^2 z \frac{\partial^2 z}{\partial x^2} = 0 \\ & -\frac{z}{x} \cdot \frac{\partial z}{\partial x} + \left(\frac{\partial z}{\partial x} \right)^2 + z \frac{\partial^2 z}{\partial x^2} = 0 \quad \dots \quad (7) \end{aligned}$$

Similarly, from (3) and (5), we get

$$-\frac{z}{y} \cdot \frac{\partial z}{\partial y} + \left(\frac{\partial z}{\partial y} \right)^2 + z \frac{\partial^2 z}{\partial y^2} = 0 \quad \dots \quad (8)$$

Finally, differentiating (2) w.r.t. y , we get

$$\begin{aligned} & 0 + a^2 \left\{ \frac{\partial z}{\partial y} \left(\frac{\partial z}{\partial x} \right) + z \frac{\partial^2 z}{\partial x \partial y} \right\} = 0 \\ \text{or,} \quad & \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + z \frac{\partial^2 z}{\partial x \partial y} = 0 \quad \dots \quad (9) \end{aligned}$$

Hence (7), (8) and (9) are three possible PDEs.

Rule II [By Eliminating Arbitrary Functions]

To find a PDE by eliminating an arbitrary function ϕ from the equation $\phi(u, v) = 0$, where u and v are functions of x, y and z .

Proof

Given equation is

$$\phi(u, v) = 0 \quad \dots \quad (1)$$

where, u and v are functions of x and y .

We assume z as dependent variable and x and y as independent variables so that

$$\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial y}{\partial x} = 0, \frac{\partial x}{\partial y} = 0 \quad \dots \quad (2)$$

Differentiating (1) partially w.r.t. x , we get

$$\begin{aligned} & \frac{\partial \phi}{\partial u} \left[\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \right] + \frac{\partial \phi}{\partial v} \left[\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial v}{\partial z} \cdot \frac{\partial z}{\partial x} \right] = 0 \\ \text{or,} \quad & \frac{\partial \phi}{\partial u} \left[\frac{\partial u}{\partial x} + 0 + \frac{\partial u}{\partial z} \cdot p \right] + \frac{\partial \phi}{\partial v} \left[\frac{\partial v}{\partial x} + 0 + \frac{\partial v}{\partial z} \cdot p \right] = 0 \\ \text{or,} \quad & \frac{\partial \phi}{\partial u} \left[\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right] + \frac{\partial \phi}{\partial v} \left[\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right] = 0 \\ & \rightarrow \frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{\left[\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right]}{\left[\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right]} \quad \dots \quad (3) \end{aligned}$$

Similarly, differentiating (1) partially with respect to y , we get

$$\rightarrow \frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{\left[\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right]}{\left[\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right]} \quad \dots \quad (4)$$

Eliminating ϕ with the help of (3) and (4), we get

$$\frac{\left[\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right]}{\left[\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right]} = \frac{\left[\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right]}{\left[\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right]}$$

$$\left[\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right] \left[\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right] = \left[\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right] \left[\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right]$$

$$Pp + Qq = R \quad \dots \quad (5)$$

where,

$$P = \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y}, \quad Q = \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} \quad \text{and} \quad R = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}$$

Thus, we obtain a linear PDE of first order and first degree in p and q.

Note

If the given equation between x, y, z containing more than one arbitrary functions, then in general, their elimination gives rise to partial differential equation of higher order.

Examples

- Form a PDE by eliminating the arbitrary function ϕ from $\phi(x + y + z, x^2 + y^2 - z^2) = 0$.
What is the degree of this PDE?

Solution

The given equation is

$$\phi(x + y + z, x^2 + y^2 - z^2) = 0 \quad \dots \quad (1)$$

$$\text{Let } u = x + y + z \text{ and } v = x^2 + y^2 - z^2 \quad \dots \quad (2)$$

Then (1) reduces to

$$\phi(u, v) = 0 \quad \dots \quad (3)$$

Differentiating (3) partially w.r.t. we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \quad \dots \quad (4)$$

From (2), we have

$$\frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial z} = 1, \frac{\partial v}{\partial x} = 2x, \frac{\partial v}{\partial z} = -2z, \frac{\partial u}{\partial y} = 1, \frac{\partial v}{\partial y} = 2y \quad \dots \quad (5)$$

From (4) and (5), we get

$$\frac{\partial \phi}{\partial u} (1 + p) + 2 \frac{\partial \phi}{\partial v} (x - pz) = 0$$

$$\frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = - \frac{2(x - pz)}{1 + p} \quad \dots \quad (6)$$

Similarly, differentiating (3) partially w.r.t. y , we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0$$

$$\frac{\partial \phi}{\partial u} (1 + q) + 2 \frac{\partial \phi}{\partial v} (y - qz) = 0, \quad \text{from (5)}$$

$$\frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = - \frac{2(y - qz)}{1 + q} \quad \dots \quad (7)$$

Eliminating ϕ from (6) and (7), we get

$$\frac{(x - pz)}{1 + p} = \frac{(y - qz)}{1 + q}$$

$$\text{or, } (1 + q)(x - pz) = (1 + p)(y - qz)$$

$$\text{or, } (y + z)p - (x + z)q = x - y$$

Hence, the required PDE of the first order is

$$(y + z)p - (x + z)q = x - y.$$

2. Form a PDE by eliminating the arbitrary function f from the equation

$$x + y + z = f(x^2 + y^2 + z^2).$$

Solution

The given equation is

$$x + y + z = f(x^2 + y^2 + z^2) \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t x and y , respectively, we get

$$1 + p = f'(x^2 + y^2 + z^2) \cdot (2x + 2zp) \quad \dots \quad (2)$$

and

$$1 + q = f'(x^2 + y^2 + z^2) \cdot (2y + 2zq) \quad \dots \quad (3)$$

Eliminating $f'(x^2 + y^2 + z^2)$ from (2) and (3), we get

$$\begin{aligned} \frac{1+p}{1+q} &= \frac{2x+2zp}{2y+2zq} \\ \text{or, } \frac{1+p}{1+q} &= \frac{2x+2zp}{2y+2zq} = \frac{x+zp}{y+zq} \\ \text{or, } (1+p)(y+zq) &= (1+q)(x+zp) \\ \text{or, } (y-z)p + (z-x)q &= x-y \end{aligned}$$

Hence, the required PDE of first order is

$$(y-z)p + (z-x)q = x-y.$$

3. Form a pde by eliminating the arbitrary functions f and F from $z = f(x + iy) + F(x - iy)$, where $i^2 = -1$.

Solution

The given equation is

$$z = f(x + iy) + F(x - iy) \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t x and y , respectively, we get

$$p = \frac{\partial z}{\partial x} = f'(x + iy) + F'(x - iy) \quad \dots \quad (2)$$

and

$$q = \frac{\partial z}{\partial y} = if'(x + iy) - iF'(x - iy) \quad \dots \quad (3)$$

Again, differentiating (2) w.r.t x and (3) with respect to y partially, we get

$$\frac{\partial^2 z}{\partial x^2} = r = f''(x+iy) + F''(x-iy) \quad \dots \quad (4)$$

And

$$\begin{aligned} \frac{\partial^2 z}{\partial y^2} &= t = i^2 f''(x+iy) + i^2 F''(x-iy) \\ \text{i.e. } \frac{\partial^2 z}{\partial y^2} &= t = -[f''(x+iy) + F''(x-iy)] \quad \dots \quad (5) \end{aligned}$$

Adding (4) and (5), we get

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = r + t = 0$$

Hence, the required PDE is

$$r + t = 0.$$

4. Form a PDE by eliminating the arbitrary function f from

$$z = e^{ax+by} f(ax - by).$$

Solution

The given equation is

$$z = e^{ax+by} f(ax - by) \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t x and y , respectively, we get

$$\begin{aligned} p &= \frac{\partial z}{\partial x} = ae^{ax+by} f(ax - by) + ae^{ax+by} f'(ax - by) \\ \Rightarrow p &= ae^{ax+by} [f(ax - by) + f'(ax - by)] \quad \dots \quad (2) \end{aligned}$$

and

$$q = be^{ax+by} [f(ax - by) - f'(ax - by)] \quad \dots \quad (3)$$

Multiplying (2) by b and (3) by a and then by addition, we get

$$\begin{aligned} bp + aq &= [abe^{ax+by} \{f(ax - by) + f'(ax - by)\} + abe^{ax+by} \{f(ax - by) - f'(ax - by)\}] \\ bp + aq &= abe^{ax+by} [f(ax - by) + f'(ax - by) + f(ax - by) - f'(ax - by)] \\ bp + aq &= 2abe^{ax+by} f(ax - by) \\ bp + aq &= 2abz, \text{ from (1)} \end{aligned}$$

Hence, the required PDE of the first order is

$$bp + aq = 2abz.$$

5. Find the PDE by eliminating the arbitrary functions f and g from $z = yf(x) + xg(y)$.

Solutoin

The given equation is

$$z = yf(x) + xg(y) \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t. x and y respectively, we get

$$p = \frac{\partial z}{\partial x} = yf'(x) + g(y) \quad \dots \quad (2)$$

and

$$q = \frac{\partial z}{\partial y} = f(x) + xg'(y) \quad \dots \quad (3)$$

Again differentiating (3) partially, w.r.t x , we get

$$s = \frac{\partial^2 z}{\partial x^2} = f'(x) + g'(y) \quad \dots \quad (4)$$

From (2) and (3), we get

$$f'(x) = \frac{1}{y} [p - g(y)]$$

And

$$g'(y) = \frac{1}{x} [q - f(x)]$$

Putting these values of $f'(x)$ and $g'(y)$ in (4), we get

$$\begin{aligned} s &= \frac{1}{y} [p - g(y)] + \frac{1}{x} [q - f(x)] \\ xys &= x[p - g(y)] + y[q - f(x)] \\ xys &= xp - xg(y) + yq - yf(x) \\ xys &= xp + yq - \{yf(x) + xg(y)\} \\ xys &= xp + yq - z, \text{ using (1)} \end{aligned}$$

Hence, the required PDE of the second order is

$$xys = xp + yq - z.$$

Linear PDE of First Order

Lagrange's Equation

A linear PDE of first order is of the form

$$Pp + wq = R$$

where P, Q and R are functions of x, y, z and p, q have their usual meaning. Such an equation is known as Lagrange's equation. For example, $xyp + yzq = zx$ is a Lagrange's equation.

Lagrange's Method of Solving $Pp + Qq = R$

Theorem

The general solution of Lagrange's equation

$$Pp + Qq = R \quad \dots \quad (1)$$

is

$$\phi(u, v) = 0 \quad \dots \quad (2)$$

where ϕ is an arbitrary function and

$$u(x, y, z) = c_1, \text{ and } v(x, y, z) = c_2 \quad \dots \quad (3)$$

are two independent solutions of the simultaneous equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots \quad (4)$$

Here, c_1 and c_2 are arbitrary constants and at least one of u, v must contain z . Also, u and v are said to be independent if $\frac{u}{v}$ is not merely a constant.

Proof

Differentiating equation (2) partially w.r.t x and y , we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \quad \dots \quad (5)$$

and

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0 \quad \dots \quad (6)$$

Eliminating $\frac{\partial \phi}{\partial u}$ and $\frac{\partial \phi}{\partial v}$ between (5) and (6), we get

$$\begin{aligned} & \begin{vmatrix} \frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} & \frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} & \frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \end{vmatrix} = 0 \\ \text{or, } & \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) - \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \\ \text{or, } & \left(\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} \right) p + \left(\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} \right) q + \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} = 0 \\ & \left(\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y} \right) p + \left(\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} \right) q = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} \quad \dots \quad (7) \end{aligned}$$

This shows that (2) is the solution of equation (7).

Next, taking the total differentials of

$u(x, y, z) = c_1$, and $v(x, y, z) = c_2$, we get

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = 0 \quad \dots \quad (8)$$

$$\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz = 0 \quad \dots \quad (9)$$

Solving for dx, dy and dz , from (8) and (9) we get

$$\frac{dx}{\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y}} = \frac{dy}{\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z}} = \frac{dz}{\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}} \quad \dots \quad (10)$$

Dividing (4) by (10), we get

$$\frac{\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y}}{P} = \frac{\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z}}{Q} = \frac{\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}}{R} = k, \text{ say}$$

Therefore,

$$\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y} = kP,$$

$$\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} = kQ$$

and

$$\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} = kR$$

Substituting these values in (7), we get

$$kPp + kQq = kR$$

$$\Rightarrow Pp + Qq = R,$$

which is the given equation (1).

This completes the proof of the theorem.

Note

The equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

are called Lagrange's Auxiliary (Subsidiary) equations of

$$Pp + Qq = R.$$

Working Rule for Solving $Pp + Qq = R$

Step 1

Put the given linear PDE of first order in the standard form

$$Pp + Qq = R \quad \dots (1)$$

Step 2

Write down Lagrange's Auxiliary equations of (1) i.e.

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots (2)$$

Step 3

Solve (2) using the methods discussed earlier (in solving ordinary differential equation). Let $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ be two independent solutions of (2).

Step 4

The general solution of (1) is then written in any one of the following forms:

$\phi(u, v) = 0$ or $u = \phi(v)$ or $v = \phi(u)$, where ϕ is an arbitrary function.

Four Rules for Getting Two Independent Solutions of

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \text{Corresponding to Four Types of } Pp + Qq = R$$

1. Rule I Corresponding to Type I

When one of the variables is either absent or cancels out from any two fractions of the given equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Examples

1. Solve $\frac{y^2 z}{x} p + xzq = y^2$

Solution

The given equation is

$$\frac{y^2 z}{x} p + xzq = y^2 \quad \dots (1)$$

The Lagrange's Auxiliary equations of (1) are

$$\frac{\frac{dx}{y^2z}}{\frac{1}{x}} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \dots \quad (2)$$

From first two ratios,

$$\begin{aligned} \frac{\frac{dx}{y^2z}}{\frac{1}{x}} &= \frac{dy}{xz} \\ \rightarrow \frac{xdx}{y^2z} &= \frac{dy}{xz} \\ \rightarrow x^2dx &= y^2dy \\ \rightarrow 3x^2dx &= 3y^2dy \end{aligned}$$

Integration implies

$$x^3 - y^3 = c_1$$

where c_1 is arbitrary constant.

Next, taking first and last ratios, we get

$$\begin{aligned} \frac{\frac{dx}{y^2z}}{\frac{1}{x}} &= \frac{dz}{y^2} \\ xdx &= zdz \\ 2xdx - 2zdz &= 0 \end{aligned}$$

Integration implies

$$x^2 - z^2 = c_2$$

where c_2 is an arbitrary constant.

Hence, the required general solution is

$$\phi(x^3 - y^3, x^2 - y^2) = 0,$$

where ϕ is an arbitrary constant.

2. Solve $y^2p - xyq = x(z - 2y)$

Solution

The given equation is

$$y^2p - xyq = x(z - 2y) \quad \dots \quad (1)$$

Lagrange's Auxiliary equations are

$$\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z - 2y)} \quad \dots \quad (2)$$

From first two ratios of (2), we have

$$\begin{aligned} \frac{dx}{y^2} &= \frac{dy}{-xy} \\ \rightarrow \frac{dx}{y} &= \frac{dy}{-x} \\ \rightarrow xdx + ydy &= 0 \\ \rightarrow 2xdx + 2ydy &= 0 \end{aligned}$$

Integration implies

$$x^2 + y^2 = c_1 \quad \dots \quad (3)$$

where c_1 is an arbitrary constant.

Next, taking last two ratios, we get

$$\begin{aligned}
\frac{dy}{-xy} &= \frac{dz}{x(z-2y)} \\
\rightarrow \frac{dy}{-y} &= \frac{dz}{z-2y} \\
\rightarrow \frac{dy}{dz} &= \frac{-y}{z-2y} \\
\rightarrow \frac{dz}{dy} &= \frac{z-2y}{-y} \\
\rightarrow \frac{dz}{dy} + \frac{1}{y}z &= 2 \quad \dots \quad (4)
\end{aligned}$$

which is linear in z with $p = \frac{1}{y}$.

Therefore,

$$I.F. = e^{\frac{1}{y}dy} = e^{\ln y} = y.$$

Multiplying (4) by $I.F. = y$, we get

$$\begin{aligned}
y \frac{dz}{dy} + z &= 2y \\
\rightarrow yz &= 2 \int y dy \\
\rightarrow yz &= y^2 + c_2 \\
\rightarrow yz - y^2 &= c_2 \quad \dots \quad (5)
\end{aligned}$$

where c_2 is an arbitrary constant.

Hence, the required general solution of (1) is

$$\phi(x^2 + y^2, yz - y^2) = 0,$$

where, ϕ is an arbitrary function.

Rule II : Corresponding to type II

Suppose that one integral of

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

is known by rule I and suppose also that another integral cannot be obtained by this rule. Then, one integral known from rule I is used to find another integral.

Examples

1. Solve $p + 3q = 5z + \tan(y - 3x)$

Solution

The given equation is

$$p + 3q = 5z + \tan(y - 3x) \quad \dots \quad (1)$$

The Lagrange's Auxiliary equation of (1) are

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)} \quad \dots \quad (2)$$

From first two ratios, we have

$$\begin{aligned}
\frac{dx}{1} &= \frac{dy}{3} \\
\rightarrow 3dx &= dy \\
\rightarrow dy - 3dx &= 0
\end{aligned}$$

Integration implies

$$y - 3x = c_1 \quad \dots \quad (3)$$

where c_1 is an arbitrary constant.

From first and last ratios we have

$$\begin{aligned}\frac{dx}{1} &= \frac{dz}{5z + \tan(y - 3x)} \\ \rightarrow \frac{dx}{1} &= \frac{dz}{5z + \tan c_1}, \quad \text{using (3)} \\ \rightarrow dx - \frac{1}{5} \left(\frac{5dz}{5z + \tan c_1} \right) &= 0\end{aligned}$$

Integration implies

$$\begin{aligned}x - \frac{1}{5} \ln(5z + \tan c_1) &= \frac{1}{5} c_2 \\ \text{or, } 5x - \ln\{5z + \tan(y - 3x)\} &= c_2 \quad \dots \quad (4)\end{aligned}$$

From (3) and (4), the required general solution is

$$5x - \ln\{5z + \tan(y - 3x)\} = \phi(y - 3x)$$

where, ϕ is an arbitrary function.

2. Solve $py + qx = xyz^2(x^2 - y^2)$

Solution

The given equation is

$$py + qx = xyz^2(x^2 - y^2) \quad \dots \quad (1)$$

Lagrange's Auxiliary equations of (1) are

$$\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{xyz^2(x^2 - y^2)} \quad \dots \quad (2)$$

From first two ratios, we have

$$\begin{aligned}\frac{dx}{y} &= \frac{dy}{x} \\ \rightarrow xdx &= ydy \\ \rightarrow xdx - ydy &= 0\end{aligned}$$

Integration implies

$$x^2 - y^2 = c_1 \quad \dots \quad (3)$$

where, c_1 is an arbitrary constant.

Taking last two ratios, we get

$$\begin{aligned}\frac{dy}{x} &= \frac{dz}{xyz^2(x^2 - y^2)} \\ \rightarrow dy &= \frac{dz}{yz^2 c_1}, \quad \text{using (3)} \\ \rightarrow ydy &= \frac{dz}{z^2 c_1} \\ \rightarrow c_1 ydy - z^{-2} dz &= 0 \\ \rightarrow 2c_1 ydy - 2z^{-2} dz &= 0\end{aligned}$$

Integration implies

$$\begin{aligned}c_1 y^2 + \frac{2}{z} &= c_2 \\ \rightarrow (x^2 - y^2)y^2 + \frac{2}{z} &= c_2, \text{using (3)} \quad \dots \quad (4)\end{aligned}$$

where c_2 is an arbitrary constant.

Hence, from (3) and (4) the required general solution is

$$y^2(x^2 - y^2) + \frac{2}{z} = \phi(x^2 - y^2)$$

where, ϕ is an arbitrary function.

Rule III Corresponding to type III

Let P_1, Q_1 and R_1 be functions of x, y and z . Then, by a well-known principle of algebra, each fraction of

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

is equal to

$$\frac{P_1 dx + Q_1 dy + R_1 dz}{P_1 P + Q_1 Q + R_1 R}.$$

If $P_1 P + Q_1 Q + R_1 R = 0$, then we know that $P_1 dx + Q_1 dy + R_1 dz = 0$, which when integrated gives $u(x, y, z) = c_1$. This method may be repeated to find another integral $v(x, y, z) = c_2$.

Here, P_1, Q_1 and R_1 are called the multipliers. As special case P_1, Q_1 and R_1 can also be constants. Sometimes, only one integral is possible by using multipliers. In such cases, the previous rules I or II are used to find another integral.

Examples

1. Solve $\left\{\frac{(b-c)}{a}\right\} yzp + \left\{\frac{(c-a)}{b}\right\} zxq = \left\{\frac{(a-b)}{c}\right\} xy$.

Solution

The given equation is

$$\left\{\frac{(b-c)}{a}\right\} yzp + \left\{\frac{(c-a)}{b}\right\} zxq = \left\{\frac{(a-b)}{c}\right\} xy \quad \dots \quad (1)$$

Lagrange's Auxiliary equation of (1) are

$$\frac{adx}{(b-c)yz} = \frac{bdy}{(c-a)zx} = \frac{cdz}{(a-b)xy} \quad \dots \quad (2)$$

Choose x, y, z as multipliers. Then, each ratio of (2) equals to

$$\begin{aligned} \frac{axdx + bydy + czdz}{xyz(b-c) + xyz(c-a) + xyz(a-b)} &= \frac{axdx + bydy + czdz}{0} \\ \therefore axdx + bydy + czdz &= 0 \\ \Rightarrow 2axdx + 2bydy + 2czdz &= 0 \end{aligned}$$

Integration implies

$$ax^2 + by^2 + cz^2 = c_1 \quad \dots \quad (3)$$

where c_1 is an arbitrary constant.

Again, let ax, by, cz be multipliers. Then, each ratio of (2) equals to

$$\begin{aligned} \frac{a^2 x dx + b^2 y dy + c^2 z dz}{xyz(ab-ca) + xyz(bc-ab) + xyz(ca-bc)} &= \frac{a^2 x dx + b^2 y dy + c^2 z dz}{0} \\ \therefore a^2 x dx + b^2 y dy + c^2 z dz &= 0 \\ \rightarrow 2a^2 x dx + 2b^2 y dy + 2c^2 z dz &= 0 \end{aligned}$$

Integration implies

$$a^2 x^2 + b^2 y^2 + c^2 z^2 = c_2 \quad \dots \quad (4)$$

where c_2 is an arbitrary constant.

From (3) and (4), the required general solution is

$$\phi(ax^2 + by^2 + cz^2, a^2 x^2 + b^2 y^2 + c^2 z^2) = 0$$

where ϕ is an arbitrary function.

2. Solve $z(x+y)p + z(x-y)q = x^2 + y^2$

Solution

The given equation is

$$z(x+y)p + z(x-y)q = x^2 + y^2 \quad \dots \quad (1)$$

Lagrange's Auxiliary equations are

$$\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2 + y^2} \quad \dots \quad (2)$$

Let $y, x, -z$ be multipliers. Then, each ratio of (2) is equal to

$$\begin{aligned} \frac{ydx + xdy - zdz}{yz(x+y) + zx(x-y) - z(x^2 + y^2)} &= \frac{ydx + xdy - zdz}{0} \\ \therefore ydx + xdy - zdz &= 0 \\ \Rightarrow d(xy) - zdz &= 0 \end{aligned}$$

Integration implies

$$\begin{aligned} xy - \frac{z^2}{2} &= c \\ \rightarrow 2xy - z^2 &= c_1 \quad \dots \quad (3) \end{aligned}$$

where c_1 is an arbitrary constant

Again, choose $x, -y, -z$ as multipliers. Then, each ratio of (2) is equal to

$$\begin{aligned} \frac{xdx - ydy - zdz}{zx(x+y) - yz(x-y) - z(x^2 + y^2)} &= \frac{xdx - ydy - zdz}{0} \\ \therefore xdx - ydy - zdz &= 0 \\ \rightarrow 2xdx - 2ydy - 2zdz &= 0 \end{aligned}$$

Integration implies

$$x^2 - y^2 - z^2 = c_2$$

where c_2 is an arbitrary constant.

Hence, from (3) and (4) the required general solution is

$$\phi(x^2 - y^2 - z^2, 2xy - z^2) = 0,$$

where ϕ is an arbitrary function.

3. Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$

Solution

The given equation is

$$x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2) \quad \dots \quad (1)$$

The Lagrange's Auxiliary equations are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)} \quad \dots \quad (2)$$

Choose $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ as multipliers. Then, each ratio of (2) equals to

$$\begin{aligned} \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{(y^2 + z) - (x^2 + z) + (x^2 - y^2)} &= \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0} \\ \therefore \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz &= 0 \end{aligned}$$

Integration implies

$$\ln x + \ln y + \ln z = \ln c_1 \quad \dots \quad (3)$$

where c_1 is an arbitrary constant.

Next, choose $x, y, -z$ as multipliers. Each ratio of (2) equals to

$$\frac{xdx + ydy - zdz}{x^2(y^2 + z) - y^2(x^2 + z) - z^2(x^2 - y^2)} = \frac{xdx + ydy - zdz}{0}$$

$$\therefore xdx + ydy - zdz = 0$$

Integration implies

$$x^2 + y^2 - z^2 = c_2 \quad \dots \quad (4)$$

where c_2 is an arbitrary constant.

Now, from (3) and (4), the required general solution of (1) is

$$\phi(x + y + z, x^2 + y^2 - z^2) = 0$$

where ϕ is an arbitrary function.

4. Solve $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$

Solution

The given equation is

$$(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx \quad \dots \quad (1)$$

The Lagrange's equation of (1) are

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx} \quad \dots \quad (2)$$

From last two ratios of (2), we get

$$\begin{aligned} \frac{dy}{xy + zx} &= \frac{dz}{xy - zx} \\ \Rightarrow \frac{dy}{y + z} &= \frac{dz}{y - z} \\ \Rightarrow (y - z)dy &= (y + z)dz \\ \Rightarrow ydy - zdy &= ydz + zdz \\ \Rightarrow ydy - zdy - (ydz + zdy) &= 0 \\ \Rightarrow ydy - zdy - d(yz) &= 0 \end{aligned}$$

Integration implies

$$y^2 - z^2 - 2yz = c_1 \quad \dots \quad (3)$$

where c_1 is an arbitrary constant

Next, choosing x, y, z as multipliers, each ratio of (2) equals to

$$\frac{xdx + ydy + zdz}{x(z^2 - 2yz - y^2) + y(xy + zx) + z(xy - zx)} = \frac{xdx + ydy + zdz}{0}$$

$$\therefore xdx + ydy + zdz = 0$$

Integration implies

$$x^2 + y^2 + z^2 = c_2 \quad \dots \quad (4)$$

where c_2 is an arbitrary constant.

Hence, the required general solution of (1) is

$$\phi(y^2 - z^2 - 2yz, x^2 + y^2 + z^2) = 0$$

where, ϕ is an arbitrary function.

5. Solve $(3x + y - z)p + (x + y - z)q = 2(z - y)$

Solution

The given equation is

$$(3x + y - z)p + (x + y - z)q = 2(z - y) \quad \dots \quad (1)$$

Lagrange's Auxiliary equations of (1) are

$$\frac{dx}{3x + y - z} = \frac{dy}{x + y - z} = \frac{dz}{2(z - y)} \quad \dots \quad (2)$$

Take 1, -3, -1 as multipliers. Then, each ratio of (2) is equal to

$$\frac{dx - 3dy - dz}{3x + y - z - 3(x + y - z) - 2(z - y)} = \frac{dx - 3dy - dz}{0}$$

$$\therefore dx - 3dy - dz = 0$$

Integration implies

$$x - 3y - z = c_1$$

$$\Rightarrow z = c_1 - x + 3y \quad \dots \quad (3)$$

where, c_1 is an arbitrary constant.

From first two ratios of (2), we get

$$\frac{dx}{3x + y - z} = \frac{dy}{x + y - z}$$

$$\Rightarrow \frac{dx}{3x + y - (c_1 - x + 3y)} = \frac{dy}{x + y - (c_1 - x + 3y)}$$

$$\Rightarrow \frac{dx}{2x + 4y + c_1} = \frac{dy}{4y + c_1} \quad \dots \quad (4)$$

Put $u = 4y + c_1$

$$\Rightarrow du = 4dy$$

$$\therefore dy = \frac{1}{4} du \quad \dots \quad (5)$$

Now, (4) implies

$$\frac{dx}{2x + u} = \frac{\frac{1}{4} du}{u}$$

$$\Rightarrow \frac{dx}{du} = \frac{1}{4} \frac{(2x + u)}{u}$$

$$\Rightarrow \frac{dx}{du} - \frac{1}{2u} x = \frac{1}{4} \quad \dots \quad (6)$$

which is linear.

So, I.F. of (6) is

$$I.F. = e^{\int -\frac{1}{2u} du} = e^{-\frac{1}{2} \int \frac{1}{u} du} = e^{-\frac{1}{2} \ln u} = e^{\ln u^{-\frac{1}{2}}} = e^{\ln \frac{1}{\sqrt{u}}} = \frac{1}{\sqrt{u}}$$

Multiplying (6) by $\frac{1}{\sqrt{u}}$, we get

$$\frac{1}{\sqrt{u}} \frac{dx}{du} - \frac{1}{\sqrt{u}} \cdot \frac{1}{2u} x = \frac{1}{4} \frac{1}{\sqrt{u}}$$

$$\Rightarrow \frac{d}{du} \left(x \cdot \frac{1}{\sqrt{u}} \right) = \frac{1}{4} \frac{1}{\sqrt{u}}$$

$$\Rightarrow \left(x \cdot \frac{1}{\sqrt{u}} \right) = \frac{1}{4} \int \frac{1}{\sqrt{u}} du$$

$$\Rightarrow x \cdot \frac{1}{\sqrt{u}} = \frac{1}{2} \sqrt{u} + c_2$$

$$\Rightarrow \frac{2x - u}{\sqrt{u}} = c_2$$

$$\Rightarrow \frac{2x - (4y + c_1)}{\sqrt{4y + c_1}} = c_2$$

$$\Rightarrow \frac{2x - 4y - (x - 3y - z)}{\sqrt{4y + x - 3y - z}} = c_2$$

$$\Rightarrow \frac{x - y + z}{\sqrt{x + y - z}} = c_2 \quad \dots \quad (7)$$

Hence, from (3) and (7), the required general solution of (1) is

$$\phi\left(x - 3y - z, \frac{x - y + z}{\sqrt{x + y - z}}\right) = 0,$$

where ϕ is an arbitrary function.

Rule IV Corresponding to Type IV

Let P_1, Q_1 and R_1 be functions of x, y and z . Then, by a well-known principle of algebra, each fraction of

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots \quad (1)$$

will be equal to

$$\frac{P_1 dx + Q_1 dy + R_1 dz}{PP_1 + QQ_1 + RR_1}.$$

If the numerator of this fraction is an exact differential of its denominator, then it can be combined with a suitable fraction of (1) to give an integral. This method can be repeated to get another integral. Some times only one integral is possible by this rule. In such cases, to find another integral, we use Rule I and Rule II as before.

Examples

1. Solve $(y + z)p + (z + x)q = x + y$

Solution

The given equation is

$$(y + z)p + (z + x)q = x + y \quad \dots \quad (1)$$

The Lagrange's Auxiliary equations are

$$\frac{dx}{y + z} = \frac{dy}{z + x} = \frac{dz}{x + y} \quad \dots \quad (2)$$

Let 1, -1, 0 be multipliers. Then, each ratio of (2) is equal to

$$\frac{dx - dy}{(y + z) - (z + x)} = \frac{d(x - y)}{-(x - y)}.$$

Again, let 0, 1, -1 be multipliers. Then, each ratio of (2) equals to

$$\frac{dy - dz}{(z + x) - (x + y)} = \frac{d(y - z)}{-(y - z)}.$$

Also, choosing 1, 1, 1 as multipliers, each ratio of (2), equals to

$$\frac{dx + dy + dz}{(y + z) + (z + x) + (x + y)} = \frac{d(x + y + z)}{2(x + y + z)}.$$

Therefore,

$$\frac{d(x - y)}{-(x - y)} = \frac{d(y - z)}{-(y - z)} = \frac{d(x + y + z)}{2(x + y + z)}.$$

Taking first two ratios of this, we get

$$\begin{aligned} \frac{d(x - y)}{-(x - y)} &= \frac{d(y - z)}{-(y - z)} \\ \Rightarrow \ln(x - y) &= \ln(y - z) + \ln c_1 \\ \Rightarrow \ln\left(\frac{x - y}{y - z}\right) &= \ln c_1 \\ \Rightarrow \frac{x - y}{y - z} &= c_1 \quad \dots \quad (3) \end{aligned}$$

where, c_1 is an arbitrary constant.

Taking first and last ratios, we get

$$\begin{aligned}\frac{d(x-y)}{-(x-y)} &= \frac{d(x+y+z)}{2(x+y+z)} \\ \Rightarrow \frac{2d(x-y)}{(x-y)} + \frac{d(x+y+z)}{(x+y+z)} &= 0\end{aligned}$$

Integration implies

$$\begin{aligned}2\ln(x-y) + \ln(x+y+z) &= \ln c_2 \\ \Rightarrow \ln(x-y)^2(x+y+z) &= \ln c_2 \\ \Rightarrow (x-y)^2(x+y+z) &= c_2 \quad \dots \quad (4)\end{aligned}$$

where c_2 is an arbitrary constant.

From (3) and (4) the required general solution is

$$\begin{aligned}\frac{x-y}{y-z} &= \phi[(x-y)^2(x+y+z)] \\ \Rightarrow x-y &= (y-z)\phi[(x-y)^2(x+y+z)]\end{aligned}$$

where ϕ is an arbitrary function.

$$2. \text{ Solve } y^2(x-y)p + x^2(y-x)q = z(x^2 + y^2)$$

Solution

The given equation is

$$y^2(x-y)p + x^2(y-x)q = z(x^2 + y^2) \quad \dots \quad (1)$$

Lagrange's Auxiliary equations are

$$\frac{dx}{y^2(x-y)} = \frac{dy}{x^2(y-x)} = \frac{dz}{z(x^2 + y^2)} \quad \dots \quad (2)$$

From first two ratios,

$$\begin{aligned}\frac{dx}{y^2(x-y)} &= \frac{dy}{x^2(y-x)} \\ \Rightarrow \frac{dx}{y^2} &= -\frac{dy}{x^2} \\ \Rightarrow x^2 dx + y^2 dy &= 0 \\ \Rightarrow 3x^2 dx + 3y^2 dy &= 0 \\ \Rightarrow x^3 + y^3 &= c_1 \quad \dots \quad (3)\end{aligned}$$

where c_1 is an arbitrary constant.

Taking 1, -1, 0 as multipliers, each ratio of (2) is equal to

$$\frac{dx - dy}{y^2(x-y) + x^2(y-x)} = \frac{d(x-y)}{(x-y)(x^2 + y^2)} \quad \dots \quad (4)$$

Taking (4) and third ratio of (2), we get

$$\begin{aligned}\frac{d(x-y)}{(x-y)(x^2 + y^2)} &= \frac{dz}{z(x^2 + y^2)} \\ \Rightarrow \frac{d(x-y)}{(x-y)} &= \frac{dz}{z} \\ \Rightarrow \ln \frac{x-y}{z} &= \ln c_2 \\ \Rightarrow \frac{x-y}{z} &= c_2 \quad \dots \quad (5)\end{aligned}$$

where c_2 is an arbitrary constant

From (3) and (5), the required general solution of (1) is

$$\phi(x^3 + y^3, \frac{x-y}{z}) = 0,$$

where, ϕ is an arbitrary function.

$$3. \text{ Solve } (x^2 - yz)p + (y^2 - zx)q = z^2 - xy$$

Solution

The given equation is

$$(x^2 - yz)p + (y^2 - zx)q = z^2 - xy \quad \dots \quad (1)$$

The Lagrange's Auxiliary equations are

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} \quad \dots \quad (2)$$

Choosing 1, -1, 0 as multipliers, each ratio of (2) is equal to

$$\frac{dx - dy}{x^2 - yz - (y^2 - zx)} = \frac{d(x - y)}{(x^2 - y^2) + z(x - y)} = \frac{d(x - y)}{(x - y)(x + y + z)} \quad \dots \quad (3)$$

And choosing 0, 1, -1 as multipliers, each ratio of (2) equals to

$$\frac{dy - dz}{(y^2 - zx) - (z^2 - xy)} = \frac{d(y - z)}{(y - z)(x + y + z)} \quad \dots \quad (4)$$

(3) and (4) imply

$$\begin{aligned} \frac{d(x - y)}{(x - y)(x + y + z)} &= \frac{d(y - z)}{(y - z)(x + y + z)} \\ \Rightarrow \frac{d(x - y)}{(x - y)} &= \frac{d(y - z)}{(y - z)} \\ \Rightarrow \ln(x - y) &= \ln(y - z) + \ln c_1 \\ \Rightarrow \frac{x - y}{y - z} &= c_1 \quad \dots \quad (5) \end{aligned}$$

where c_1 is an arbitrary constant.

Taking x, y, z as multipliers, we get each ratio of (2) equal to

$$\frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz} = \frac{xdx + ydy + zdz}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)} \quad \dots \quad (6)$$

Again taking 1, 1, 1 as multipliers each ratio of (2) equals to

$$\frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \quad \dots \quad (7)$$

From (6) and (7), we get

$$\begin{aligned} \frac{xdx + ydy + zdz}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)} &= \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \\ \Rightarrow \frac{xdx + ydy + zdz}{(x + y + z)} &= dx + dy + dz \\ \Rightarrow xdx + ydy + zdz &= (x + y + z)d(x + y + z) \\ \Rightarrow 2xdx + 2ydy + 2zdz &= 2(x + y + z)d(x + y + z) \end{aligned}$$

Integration implies

$$\begin{aligned} x^2 + y^2 + z^2 &= (x + y + z)^2 + c_2 \\ \Rightarrow x^2 + y^2 + z^2 - (x + y + z)^2 &= c_2 \\ \Rightarrow xy + yz + zx &= c_2 \quad \dots \quad (9) \end{aligned}$$

where, c_2 is an arbitrary constant.

From (5) and (9), the required general solution of (1) is

$$\begin{aligned} \frac{x - y}{y - z} &= \phi(xy + yz + zx) \\ \Rightarrow x - y &= (y - z)\phi(xy + yz + zx) \end{aligned}$$

where, ϕ is an arbitrary function.

4. Solve $(x^2 + y^2 + yz)p + (x^2 + y^2 - zx)q = z(x + y)$

Solution

The given equation is

$$(x^2 + y^2 + yz)p + (x^2 + y^2 - zx)q = z(x + y) \quad \dots \quad (1)$$

The Lagrange's Auxiliary equation are

$$\frac{dx}{x^2 + y^2 + yz} = \frac{dy}{x^2 + y^2 - zx} = \frac{dz}{z(x + y)} \quad \dots \quad (2)$$

Taking 1, -1, 0 as multipliers each ratio of (2) equals to

$$\frac{dx - dy}{(x^2 + y^2 + yz) - (x^2 + y^2 - zx)} = \frac{d(x - y)}{z(x + y)} \quad \dots \quad (3)$$

Taking x, y, 0 as multipliers, each ratio of (2) equals to

$$\begin{aligned} \frac{xdx + ydy}{x(x^2 + y^2 + yz) + y(x^2 + y^2 - zx)} &= \frac{xdx + ydy}{x^3 + xy^2 + xyz + x^2y + y^3 - xyz} \\ &= \frac{xdx + ydy}{(x + y)(x^2 + y^2)} \quad \dots \quad (4) \end{aligned}$$

Now, from third ratio of (2) and (3) and (4) we get

$$\begin{aligned} \frac{dz}{z(x + y)} &= \frac{d(x - y)}{z(x + y)} = \frac{xdx + ydy}{(x + y)(x^2 + y^2)} \\ \Rightarrow \frac{dz}{z} &= \frac{d(x - y)}{z} = \frac{xdx + ydy}{(x^2 + y^2)} \quad \dots \quad (5) \end{aligned}$$

Taking first two ratios of (5) we have

$$\begin{aligned} \frac{dz}{z} &= \frac{d(x - y)}{z} \\ \Rightarrow dz &= d(x - y) \end{aligned}$$

Integration implies

$$z - x + y = c_1 \quad \dots \quad (6)$$

c_1 being an arbitrary constant.

Next, taking first and third ratio of (5), we get

$$\begin{aligned} \frac{dz}{z} &= \frac{xdx + ydy}{(x^2 + y^2)} \\ \Rightarrow \frac{2dz}{z} &= \frac{2xdx + 2ydy}{(x^2 + y^2)} \\ \Rightarrow \frac{2dz}{z} &= \frac{d(x^2) + d(y^2)}{(x^2 + y^2)} = \frac{d(x^2 + y^2)}{(x^2 + y^2)} \\ \Rightarrow \frac{d(x^2 + y^2)}{(x^2 + y^2)} &= \frac{2dz}{z} \end{aligned}$$

Integration implies

$$\begin{aligned} \ln(x^2 + y^2) &= 2\ln z + \ln c_2 \\ \Rightarrow \ln(x^2 + y^2) - \ln z^2 &= \ln c_2 \\ \Rightarrow \frac{x^2 + y^2}{z^2} &= c_2 \quad \dots \quad (7) \end{aligned}$$

Hence, from (6) and (7) the required general solution of (1) is

$$\phi\left(z - x + y, \frac{x^2 + y^2}{z^2}\right) = 0$$

where, φ is an arbitrary function.

Integral Surface Passing Through a Given Curve

We have already discussed the Methods to find the general solution of $Pp + Qq = R$. We now discuss two methods of using such general solution to find the integral surface which passes through a given surface.

Method I

Let the given differential equation be

$$Pp + Qq = R \quad \dots \quad (1)$$

Let $u(x, y, z) = c_1$ and $v(x, y, z) = c_2 \quad \dots \quad (2)$

be two independent solutions of Lagrange's Auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots \quad (3)$$

Suppose we wish to find an integral surface which passes through the curve whose equation in parametric form is given by

$$x = x(t), y = y(t), z = z(t) \quad \dots \quad (4)$$

where, t is a parameter.

Then (2) can be written as

$$u[x(t), y(t), z(t)] = c_1 \text{ and } v[x(t), y(t), z(t)] = c_2 \quad \dots \quad (4)$$

We now eliminate the single parameter t between the equations of (4) to get a relation in c_1 and c_2 .

Then, we replace, c_1 and c_2 using (2) and obtain the required integral surface.

Exmaples

Find the integral surface of the linear PDE

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$$

which contains the straight lines

$$x + y = 0, z = 1.$$

Solution

The given PDE is

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z \quad \dots \quad (1)$$

Lagrange's Auxiliary equations of (1) are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{(x^2 - y^2)z} \quad \dots \quad (2)$$

Taking $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ as multipliers, each ratio of (2) equals to

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{y^2 + z - x^2 - z + x^2 - y^2} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

Therefore,

$$\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = 0$$

Integration implies

$$\ln x + \ln y + \ln z = \ln c_1 \Rightarrow xyz = c_1 \quad \dots \quad (3)$$

where c_1 is arbitrary constant.

Again, taking $x, y, -1$ as multipliers, each ratio of (2) equals to

$$\frac{xdx + ydy - dz}{x^2(y^2 + z) - y^2(x^2 + z) - (x^2 - y^2)z} = \frac{xdx + ydy - dz}{0}$$

$$\therefore xdx + ydy - dz = 0$$

Integration implies

$$x^2 + y^2 - 2z = c_2 \quad \dots \quad (4)$$

where, c_2 is an arbitrary constant.

Taking t as a parameter, the equation of the straight lines

$$x + y = 0, z = 1$$

can be written in parametric form as

$$x = t, y = -t, z = 1 \quad \dots \quad (5)$$

Using (5), equations (3) and (4) can be written as

$$-t^2 = c_1 \text{ and } 2t^2 - 2 = c_2 \quad \dots \quad (6)$$

Eliminating t from (6), we get

$$\begin{aligned} 2(-c_1) - 2 &= c_2 \\ \text{or, } 2c_1 + c_2 + 2 &= 0 \quad \dots \quad (7) \end{aligned}$$

Using equation (3) and (4) in (7) we get

$$2xyz + x^2 + y^2 - 2z + 2 = 0$$

Hence, the required integral surface is

$$2xyz + x^2 + y^2 - 2z + 2 = 0.$$

Method II

Let $u(x, y, z) = c_1$, and $v(x, y, z) = c_2 \quad \dots \quad (1)$

be independent solutions of the Lagrange's Auxiliary equations of

$$Pp + Qq = R \quad \dots \quad (2)$$

Suppose we wish to obtain the integral surface passing through the curve which is determined by

$$\phi(x, y, z) = 0 \text{ and } \psi(x, y, z) = 0 \quad \dots \quad (3)$$

First, we eliminate x, y, z from equations (1) and (3) to get a relation in c_1 and c_2 . Then, we replace c_1 and c_2 by using equation (1) to get the required integral surface.

Example

Find the integral surface of the PDE $(x - y)p + (y - x - z)q = z$ through the circle $z = 1, x^2 + y^2 = 1$.

Solution

The given PDE is

$$(x - y)p + (y - x - z)q = z \quad \dots \quad (1)$$

Lagrange's Auxiliary equations of (1)

$$\frac{dx}{x - y} = \frac{dy}{y - x - z} = \frac{dz}{z} \quad \dots \quad (2)$$

Choose 1, 1, 1 as multipliers so that each ratio of (2) equals to

$$\begin{aligned} \frac{dx + dy + dz}{x - y + y - x - z + z} &= \frac{dx + dy + dz}{0} \\ \therefore dx + dy + dz &= 0 \end{aligned}$$

Integration implies

$$x + y + z = c_1 \quad \dots \quad (3)$$

Taking last two ratios of (2), we get

$$\begin{aligned} \frac{dy}{y - x - z} &= \frac{dz}{z} \\ \Rightarrow \frac{dy}{y - (x + z)} &= \frac{dz}{z} \\ \Rightarrow \frac{dy}{y - (c_1 - y)} &= \frac{dz}{z}, \text{ using (3)} \end{aligned}$$

$$\Rightarrow \frac{dy}{2y - c_1} = \frac{dz}{z}$$

$$\Rightarrow \frac{2dy}{2y - c_1} - \frac{2dz}{z} = 0$$

Integration implies

$$\ln(2y - c_1) - 2\ln z = \ln c_2$$

$$\Rightarrow \frac{2y - c_1}{z^2} = c_2$$

$$\Rightarrow \frac{y - x - z}{z^2} = c_2 \quad \dots \quad (4)$$

The given curve is

$$z = 1, x^2 + y^2 = 1 \quad \dots \quad (5)$$

Putting $z = 1$ in (3) and (4) we get

$$x + y = c_1 - 1 \text{ and } y - x = c_2 + 1 \quad \dots \quad (6)$$

We know,

$$(x + y)^2 + (y - x)^2 = 2(x^2 + y^2) \quad \dots \quad (7)$$

Using (5) and (6) in (7), we get

$$(c_1 - 1)^2 + (c_2 + 1)^2 = 2$$

$$\Rightarrow c_1^2 + c_2^2 - 2c_1 + 2c_2 = 0 \quad \dots \quad (8)$$

Putting the values of c_1 and c_2 from (3) and (4), the equation (8) reduces to

$$(x + y + z)^2 + \frac{(y - x - z)^2}{z^4} - 2(x + y + z) + \frac{2(y - x - z)}{z^2} = 0$$

$$z^4(x + y + z)^2 + (y - x - z)^2 - 2z^4(x + y + z) + 2z^2(y - x - z) = 0$$

which is the required integral surface.

Surfaces Orthogonal to a Given System of Surfaces

Let $f(x, y, z) = c \quad \dots \quad (1)$

be a family of surfaces where c being a parameter. We wish to get a system of surfaces which cut each of (1) at right angles. Then, the direction ratio of the normal at the point (x, y, z) to (1) are $\frac{\partial f}{\partial x}$,

$\frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$.

Let $z = f(x, y) \quad \dots \quad (2)$

be the surface. Which cuts each of (1) at right angles. Then, the direction ratios of the normal at (x, y, z) to (2) are $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, -1$ i.e. $p, q, -1$. Since the normals to (1) and (2) are at right angles, we have

$$\frac{\partial f}{\partial x}p + \frac{\partial f}{\partial y}q + \frac{\partial f}{\partial z}(-1) = 0$$

$$\Rightarrow \frac{\partial f}{\partial x}p + \frac{\partial f}{\partial y}q = \frac{\partial f}{\partial z} \quad \dots \quad (3)$$

which is of the form $Pp + Qq = R$.

Conversely, each and every solution of (3) is orthogonal to each and every surface of (1).

Examples

- Find the surface which intersects the surfaces of the system $z(x + y) = c(3z + 1)$ orthogonally and which passes through the circle $x^2 + y^2 = 1, z = 1$.

Solution

The given surface is

$$f(x, y, z) = \frac{z(x+y)}{3z+1} = c \quad \dots \quad (1)$$

$$\therefore \frac{\partial f}{\partial x} = \frac{z}{3z+1}, \frac{\partial f}{\partial y} = \frac{z}{3z+1},$$

$$\frac{\partial f}{\partial z} = (x+y) \left[\frac{(3z+1) \cdot 1 - 3z}{(3z+1)^2} \right] = \frac{x+y}{(3z+1)^2}$$

The required orthogonal surface is a solution of

$$p \frac{\partial f}{\partial x} + q \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z}$$

$$\text{or,} \quad p \frac{z}{3z+1} + q \frac{z}{3z+1} = \frac{x+y}{(3z+1)^2}$$

$$\text{or,} \quad z(3z+1)p + z(3z+1)q = x+y \quad \dots \quad (2)$$

Lagrange's Auxiliary equations of (2) are

$$\frac{dx}{z(3z+1)} = \frac{dy}{z(3z+1)} = \frac{dz}{x+y} \quad \dots \quad (3)$$

Taking first two ratios, we get

$$\begin{aligned} \frac{dx}{z(3z+1)} &= \frac{dy}{z(3z+1)} \\ \Rightarrow dx &= dy \Rightarrow dx - dy = 0 \end{aligned}$$

Integration implies

$$x - y = c_1 \quad \dots \quad (4)$$

Taking $x, y, -z(3z+1)$ as multipliers, each ratio of (3) equals to

$$\begin{aligned} \frac{xdx + ydy - z(3z+1)dz}{xz(3z+1) + yz(3z+1) - z(x+y)(3z+1)} &= \frac{xdx + ydy - z(3z+1)dz}{0} \\ \therefore xdx + ydy - z(3z+1)dz &= 0 \end{aligned}$$

Integration implies

$$x^2 + y^2 - 2z^3 - z^2 = c_2 \quad \dots \quad (5)$$

where, c_1 and c_2 are arbitrary constants.

Hence, the general solution of (2) is

$$x^2 + y^2 - 2z^3 - z^2 = \phi(x-y) \quad \dots \quad (6)$$

where, ϕ is an arbitrary function, which is an orthogonal surface of to the system (1).

Next, to get the surface through the circle

$$x^2 + y^2 = 1, z = 1,$$

we put these values in (6), to get

$$\begin{aligned} 1 - 2 - 1 &= \phi(x-y) \\ \Rightarrow \phi(x-y) &= -2 \end{aligned}$$

Hence, the required particular surface orthogonal to (1) is

$$x^2 + y^2 - 2z^3 - z^2 + 2 = 0.$$

2. Find the surface which is orthogonal to the one parameter family of surfaces $z = cxy$ ($x^2 + y^2$) which passes through the hyperbola $x^2 - y^2 = a^2, z = 0$.

Solution

The given system of surfaces is

$$f(x, y, z) = \frac{z}{x^3y + xy^3} = c \quad \dots \quad (1)$$

$$\therefore \frac{\partial f}{\partial x} = \frac{-z(3x^2y + y^3)}{(x^3y + xy^3)^2} = P$$

$$\frac{\partial f}{\partial y} = \frac{-z(3y^2x + x^3)}{(x^3y + xy^3)^2} = Q$$

$$\frac{\partial f}{\partial z} = \frac{1}{(x^3y + xy^3)^2} = R$$

Therefore, the required orthogonal surface is a solution of

$$\frac{-z(3x^2y + y^3)}{(x^3y + xy^3)^2}p - \frac{z(3y^2x + x^3)}{(x^3y + xy^3)^2}q = \frac{1}{(x^3y + xy^3)^2}$$

$$\text{or, } \frac{yz(3x^2 + y^2)}{x^2y^2(x^2 + y^2)^2}p + \frac{zx(3y^2 + x^2)}{x^2y^2(x^2 + y^2)^2}q = -\frac{1}{xy(x^2 + y^2)}$$

$$\text{or, } yz(3x^2 + y^2)p + zx(3y^2 + x^2)q = -xy(x^2 + y^2)$$

$$\text{or, } yz(3x^2 + y^2)p + zx(3y^2 + x^2)q = -xy(x^2 + y^2)$$

$$\text{or, } \frac{(3x^2 + y^2)}{x}p + \frac{(3y^2 + x^2)}{y}q = -\frac{x^2 + y^2}{z} \quad \dots (2)$$

Lagrange's Auxiliary equations of (2) are:

$$\frac{dx}{(3x^2 + y^2)} = \frac{dy}{(3y^2 + x^2)} = \frac{dz}{-(x^2 + y^2)} \quad \dots (3)$$

From first two ratios of (3), we get

$$2xdx - 2ydy = 0$$

Integration implies

$$x^2 - y^2 = c_1 \quad \dots (4)$$

Taking $x, y, 4z$ as multipliers, each ratio of (3) is equal to

$$\frac{xdx + ydy + 4zdz}{0}$$

$$\therefore xdx + ydy + 4zdz = 0$$

Integration implies

$$x^2 + y^2 + 4z^2 = c_2 \quad \dots (5)$$

Hence, any surface orthogonal to (1) is

$$x^2 + y^2 + 4z^2 = \phi(x^2 - y^2) \quad \dots (6)$$

where ϕ is an arbitrary function.

Next, using $x^2 - y^2 = a^2$ and $z = 0$ in (6), we get

$$x^2 + y^2 = \phi(x^2 - y^2)$$

$$\Rightarrow \sqrt{(x^2 + y^2)^2} = \phi(x^2 - y^2)$$

$$\Rightarrow \sqrt{(x^2 - y^2)^2 + 4x^2y^2} = \phi(x^2 - y^2)$$

$$\Rightarrow \sqrt{a^4 + 4x^2y^2} = \phi(x^2 - y^2)$$

$$\therefore \phi(x^2 - y^2) = \sqrt{a^4 + 4x^2y^2}$$

Hence, the required particular orthogonal surface is

$$x^2 + y^2 + 4z^2 = \sqrt{a^4 + 4x^2y^2}$$

$$\Rightarrow (x^2 + y^2 + 4z^2)^2 = a^4 + 4x^2y^2$$

Non-linear Partial Differential Equations of First Order

Different Classes (Types) of Solution

A solution (or integral) of a differential equation is a relation between the variables which when substituted the equation is satisfied.

Complete Integral/Complete Solution

Let

$$\phi(x, y, z, a, b) = 0 \quad \dots \quad (1)$$

where x, y, z are variables such that z is dependent on x and y .

Differentiating (1) partially w.r.t. x and y respectively, we get

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z} p = 0 \text{ and } \frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z} q = 0 \quad \dots \quad (2)$$

Eliminating the arbitrary constants a and b from (1) and (2), a relation of the form

$$f(x, y, z, p, q) = 0 \quad \dots \quad (3)$$

will be obtained which is a PDE of order one.

Let us suppose that (1) is obtained from (3), then the integral (1) which has as many arbitrary constants as there are independent variables, is called the complete integral of (3).

Particular Integral / Particular Solution

A particular integral of (3) is obtained by assigning particular values to the arbitrary constants a and b .

Singular Integral / Singular Solution

The envelope of the surfaces given by (2) as obtained by eliminating a and b between the equations

$$\phi(x, y, z, a, b) = 0, \quad \frac{\partial \phi}{\partial a} = 0, \quad \frac{\partial \phi}{\partial b} = 0 \quad \dots \quad (2)$$

The relation between x, y, z so obtained is called the singular integral. It is different from the complete integral. In case of exponential functions, the singular integral is obtained by giving particular values to the arbitrary constant in the complete integral.

General Integral/ General Solution

Let us assume that one of the constants in (1) is a function of the other, say $b = F(a)$. Then, (1) reduces to

$$\phi(x, y, z, a, F(a)) = 0 \quad \dots \quad (5)$$

Now, (5) represents one of the families of surfaces given by the system (1). As before, the equation of the envelop of the family of surfaces given by (5) must also satisfy (3). Again, the equation so obtained will be distinct from that of the envelope of the surfaces, and it is known as the general integral and is obtained by eliminating a between

$$\phi(x, y, z, a, F(a)) = 0; \quad \frac{\partial \phi}{\partial a} = 0 \quad \dots \quad (6)$$

Compatible System of First Order Equations

Any two first order PDE

$$F(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

$$g(x, y, z, p, q) = 0 \quad \dots \quad (2)$$

are said to be compatible if every solution of one is also a solution of the other.

Theorem (Only Statement)

A necessary and sufficient condition that a pair of first order partial differential equations

$$f(x, y, z, p, q) = 0 \text{ and } g(x, y, z, p, q) = 0$$

is that

$$[f, g] = 0$$

provided

$$J = \frac{\partial(f, g)}{\partial(p, q)} = \begin{vmatrix} \frac{\partial f}{\partial p} & \frac{\partial f}{\partial q} \\ \frac{\partial g}{\partial p} & \frac{\partial g}{\partial q} \end{vmatrix} = \begin{vmatrix} f_p & f_q \\ g_p & g_q \end{vmatrix} = f_p g_q - f_q g_p \neq 0$$

where,

$$[f, g] = \frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)}.$$

Corollary

The first order partial differential equations

$$p = P(x, y) \text{ and } q = Q(x, y)$$

are compatible iff

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Example 1

Show that $p = x - \frac{y}{x^2+y^2}$ and $q = y + \frac{x}{x^2+y^2}$ are compatible and find their common solution.

Solution

The given system is

$$p = x - \frac{y}{x^2+y^2}; \quad q = y + \frac{x}{x^2+y^2} \quad \dots \quad (1)$$

Comparing the system with

$$p = P(x, y); \quad q = Q(x, y)$$

we have

$$p(x, y) = x - \frac{y}{x^2 + y^2} \quad \dots \quad (2)$$

$$Q(x, y) = y + \frac{x}{x^2 + y^2} \quad \dots \quad (3)$$

Therefore, from (2)

$$\begin{aligned} \frac{\partial P}{\partial y} &= 0 - \frac{1 \cdot (x^2 + y^2) - 2y \cdot y}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \\ \frac{\partial Q}{\partial x} &= 0 + \frac{1 \cdot (x^2 + y^2) - 2x \cdot x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \end{aligned}$$

Hence,

$$\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial Q}{\partial x}$$

and so, the system is compatible.

To find the common solution

Putting the value of p and q in

$$dz = p dx + q dy$$

we get

$$dz = \left\{ x - \frac{y}{x^2 + y^2} \right\} dx + \left\{ y + \frac{x}{x^2 + y^2} \right\} dy$$

Integrating, we get

$$z = \int \left(x - \frac{y}{x^2 + y^2} \right) dx \text{ (keeping } y \text{ constant)} + \int \left(y - \frac{x}{x^2 + y^2} \right) dy \text{ (free from } x)$$

$$z = \int x dx - y \int \frac{1}{x^2 + y^2} dx + \int y dy$$

$$z = \frac{x^2}{2} - \frac{y}{y} \tan^{-1} \frac{x}{y} + \frac{y^2}{2} + C$$

$$z = \frac{x^2}{2} + \frac{y^2}{2} - \tan^{-1} \frac{x}{y} + C$$

Therefore, the required common solution is

$$z = \frac{x^2 + y^2}{2} - \tan^{-1} \frac{x}{y} + C$$

Note

Given

$$\frac{\partial z}{\partial x} = p = P(x, y) \text{ and } \frac{\partial z}{\partial y} = q = Q(x, y) \quad \dots \quad (1)$$

Since

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy$$

$$i. e. dz = P(x, y) dx + Q(x, y) dy \quad \dots \quad (2)$$

It follows the given PDE (1) are compatible iff the single differential equation

$$dz = P(x, y) dx + Q(x, y) dy$$

is integrable.

since P and Q are functions of x and y , $p dx + Q dy$ is an exact differential iff $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Therefore, (2) is integrable iff

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Note

If

$$dz = P dx + Q dy$$

is integrable and

$$P dx + Q dy$$

is exact, then integration implies

$Z = \text{Integration of } P \text{ w.r.t. } x \text{ keeping } y \text{ constant} + \text{integration of the terms of } Q \text{ free form } x.$

Example 2

Show that $p = 1 + e^{\frac{x}{y}}$, $q = e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right)$ are compatible and find their solution.

Solution

Given equation are

$$p = 1 + e^{\frac{x}{y}} = P(x, y)$$

$$\text{and } q = e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) = Q(x, y) \quad \dots \quad (1)$$

Now,

$$\frac{\partial P}{\partial y} = e^{\frac{x}{y}} \left(-\frac{x}{y^2}\right) = -\frac{x}{y^2} e^{\frac{x}{y}}$$

and

$$\frac{\partial Q}{\partial x} = \frac{1}{y} e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) - \frac{1}{y} e^{\frac{x}{y}} = -\frac{x}{y^2} e^{\frac{x}{y}}$$

Here, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Hence, the system (1) is compatible.

Next, to Find the Common Solution

We put the value of p and q in

$$dz = p dx + q dy$$

$$\Rightarrow dz = \left(1 + e^{\frac{x}{y}}\right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy$$

Since the RHS exact, integration implies

$$\begin{aligned} z &= \int \left(1 + e^{\frac{x}{y}}\right) dx \\ \Rightarrow z &= x + \frac{e^{\frac{x}{y}}}{\frac{1}{y}} + C \\ \Rightarrow z &= x + ye^{\frac{x}{y}} + C \end{aligned}$$

where, C is an arbitrary constant.

Example 3

Show that the equations $xp = yq$ and $z(xp + yq) = 2xy$ are compatible and solve them.

Solution

Given equations are

$$xp = yq \Rightarrow xp - yq = 0$$

and

$$z(xp + yq) = 2xy \Rightarrow z(xp + yq) - 2xy = 0$$

Let

$$f(x, y, z, p, q) = xp - yq = 0 \quad \dots \quad (1)$$

and

$$g(x, y, z, p, q) = z(xp + yq) - 2xy = 0 \quad \dots \quad (2)$$

$$\therefore \frac{\partial(f, g)}{\partial(x, p)} = \begin{vmatrix} f_x & f_p \\ g_x & g_p \end{vmatrix} = \begin{vmatrix} p & x \\ zp - 2y & zx \end{vmatrix} = 2xy$$

$$\frac{\partial(f, g)}{\partial(z, p)} = \begin{vmatrix} f_z & f_p \\ g_z & g_p \end{vmatrix} = \begin{vmatrix} 0 & x \\ xp + yq & zx \end{vmatrix} = -x^2p - xyq$$

$$\frac{\partial(f, g)}{\partial(y, q)} = \begin{vmatrix} f_y & f_q \\ g_y & g_q \end{vmatrix} = \begin{vmatrix} -q & -y \\ zq - 2x & yz \end{vmatrix} = -2xy$$

$$\frac{\partial(f, g)}{\partial(z, q)} = \begin{vmatrix} f_z & f_q \\ g_z & g_q \end{vmatrix} = \begin{vmatrix} 0 & -y \\ xp + yq & yz \end{vmatrix} = xyp + y^2q$$

$$\begin{aligned} \therefore [f, g] &= \frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)} \\ &= 2xy - x^2p^2 - xypq - 2xy + xypq + y^2q^2 \\ &= -xp(xp + yq) + yq(xp + yq) \\ &= (xp - yq)(xp + yq) = 0, \text{ using (1)} \end{aligned}$$

Hence, (1) and (2) are compatible.

Next, solving (1) and (2) for p and q we get $p = \frac{y}{z}, q = \frac{x}{z}$.

Using these values in

$$dz = p dx + q dy$$

we get

$$\begin{aligned} dz &= \frac{y}{z} dx + \frac{x}{z} dy \\ \Rightarrow z dz &= y dx + x dy \\ \Rightarrow z dz &= d(xy) \end{aligned}$$

Integration Integration implies

$$z^2 = 2xy + C$$

where, C is an arbitrary constant.

Hence, the required common solution of (1) and (2) is

$$z^2 = 2xy + C.$$

Charpit's Method

General Method of Solving PDE of Order One but of Any Degree

Let the given non-linear PDE of order one be

$$f(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

We know

$$dz = p dx + q dy \quad \dots \quad (2)$$

We need to find another relation

$$F(x, y, z, p, q) = 0 \quad \dots \quad (3)$$

so that when the values of p and q obtained by solving (1) and (3) are substituted in (2) it becomes integrable. The integration of (2) then gives the complete integral of (1).

Differentiating (1) and (3) partially with respect to x and y we get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = 0 \quad \dots \quad (4)$$

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial F}{\partial q} \cdot \frac{\partial q}{\partial x} = 0 \quad \dots \quad (5)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y} = 0 \quad \dots \quad (6)$$

$$\frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} q + \frac{\partial F}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial F}{\partial q} \cdot \frac{\partial q}{\partial y} = 0 \quad \dots \quad (7)$$

Eliminating $\frac{\partial p}{\partial x}$ from (4) and (5), we get

$$\begin{aligned} & \left[\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} \right] \frac{\partial F}{\partial p} - \left[\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial q} \cdot \frac{\partial q}{\partial x} \right] \frac{\partial f}{\partial p} = 0 \\ \Rightarrow & \left[\frac{\partial f}{\partial x} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial x} \cdot \frac{\partial f}{\partial p} \right] + \left[\frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial p} \right] p + \left[\frac{\partial f}{\partial q} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial q} \cdot \frac{\partial f}{\partial p} \right] \frac{\partial q}{\partial x} = 0 \quad \dots \quad (8) \end{aligned}$$

Similarly eliminating $\frac{\partial q}{\partial y}$ from (6) and (7), we get

$$\left[\frac{\partial f}{\partial y} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial y} \cdot \frac{\partial f}{\partial q} \right] + \left[\frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial q} \right] q + \left[\frac{\partial f}{\partial p} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial p} \cdot \frac{\partial f}{\partial q} \right] \frac{\partial p}{\partial y} = 0 \quad \dots \quad (9)$$

The expression in the bracket of last terms of (8) and (9) are equal but of opposite sign and

$$\frac{\partial q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y}.$$

So, on adding of (8) and (9) those terms cancel out. Therefore, adding (8) and (9) we get

$$\left(\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial p} + \left(\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial q} + \left(-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial z} + \left(-\frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial y} = 0 \quad \dots \quad (10)$$

This is a linear equation of first order to obtain the desired function F . The integral of (1) is obtained by solving the following simultaneous equations.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0} \quad \dots \quad (11)$$

Or

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0} \quad \dots \quad (12)$$

The equations in (11) or in (12) are called the Charpit's Auxiliary equations.

Since any of the integrals of (11) (or 12) will satisfy (10), an integral of (11), which involve p and q or both will help along with the given equation to find p and q . In practice, we generally select the simplest integral.

Working Rule

1. Express the given equation as a function $f(x, y, z, p, q) = 0$
2. Write down the Charpit's Auxiliary equations.
3. Using the value of f in step (1), write down the values of f_x, f_y, f_z, f_p, f_q etc. in the auxiliary equations in (2)
4. After simplifying step (3), select two proper fractions so that the resulting integral involves at least one of p and q .
5. The simplest integral obtained in step (4) and the given equation are solved to find p and q .
6. Put the values of p and q obtained in step (5) in

$$dz = p dx + q dy$$

which when integrated gives the complete integral of the given equation.

Note

Some times charpit's Auxiliary equations give $p = a$ and $q = b$, where a and b are arbitrary constants. In such cases, we put $p = a$ and $q = b$ in $dz = p dx + q dy$ and integrate to get the complete integral.

Examples

1. Find a complete integral of $z = px + qy + p^2 + q^2$.

Solution

The given equation is

$$z = px + qy + p^2 + q^2 \quad \dots \quad (1)$$

Let

$$f(x, y, z, p, q) = z - px - qy - p^2 - q^2 = 0 \quad \dots \quad (2)$$

Differentiating (2) partially w.r.t. x, y, z, p and q , we get

$$f_x = -p, f_y = -q, f_z = 1, f_p = -x - 2p, f_q = -y - 2q \quad \dots \quad (3)$$

Charpit's Auxiliary equations are

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

Using (3), we get

$$\begin{aligned} \frac{dp}{-p + p \cdot 1} &= \frac{dq}{-q + q \cdot 1} = \frac{dz}{p(x + 2p) + q(y + 2q)} = \frac{dx}{x + 2p} = \frac{dy}{y + 2q} \\ \Rightarrow \frac{dp}{0} &= \frac{dq}{0} = \frac{dz}{p(x + 2p) + q(y + 2q)} = \frac{dx}{x + 2p} = \frac{dy}{y + 2q} \quad \dots \quad (4) \end{aligned}$$

From first fraction we have

$$dp = 0 \Rightarrow p = a \quad \dots \quad (5)$$

and from second fraction we have

$$dq = 0 \Rightarrow q = b \quad \dots \quad (6)$$

Putting these values of p and q in (1), the required complete integral is

$$z = ax + by + a^2 + b^2$$

where a and b are arbitrary constant.

2. Find a complete integral of $q = 3p^2$

Solution

The given equation is

$$q = 3p^2 \quad \dots \quad (1)$$

$$\text{Let } f(x, y, z, p, q) = q - 3p^2 = 0 \quad \dots \quad (2)$$

$$\therefore f_x = 0, f_y = 0, f_z = 0, f_p = 6p, f_q = 1 \quad \dots \quad (3)$$

Charpit's Auxiliary equations are

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

Using (3), it reduces to

$$\frac{dp}{0} = \frac{dq}{0} = \frac{dz}{-6p^2 + q} = \frac{dx}{-6p} = \frac{dy}{1} \quad \dots \quad (4)$$

From first ratio of (4), we have

$$dp = 0 \Rightarrow p = a \quad \dots \quad (5)$$

Using (5) in (1), we get

$$q = 3a^2 \quad \dots \quad (6)$$

Using (5) and (6) in

$$dz = p dx + q dy$$

we get

$$dz = a dx + 3a^2 dy$$

Integration implies

$$z = ax + 3a^2 y + C$$

where, C is an arbitrary constnt.

Hence, the required complete integration is

$$z = ax + 3a^2 y + C.$$

3. Find a complete integral of $p^2 - y^2 q = y^2 - x^2$

Solution

The given equation is

$$p^2 - y^2 q = y^2 - x^2 \quad \dots \quad (1)$$

$$\text{Let } f(x, y, z, p, q) = p^2 - y^2 q - y^2 + x^2 = 0 \quad \dots \quad (2)$$

$$\therefore f_x = 2x, f_y = -2yq - 2y, f_z = 0, f_p = 2p, f_q = -y^2 \quad \dots \quad (3)$$

Charpit's Auxiliary equation are

$$\begin{aligned} \frac{dp}{f_x + pf_z} &= \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} \\ \Rightarrow \frac{dp}{2x} &= \frac{dq}{-2yq - 2y} = \frac{dz}{-2p^2 + q^3} = \frac{dx}{-2p} = \frac{dy}{y^2} \quad \dots \quad (4) \end{aligned}$$

Taking first and fourth fractions of (4) we get

$$\begin{aligned} \frac{dp}{2x} &= \frac{dx}{-2p} \\ \Rightarrow p dp + x dx &= 0 \\ \Rightarrow p^2 + x^2 &= a^2 \quad \dots \quad (5) \end{aligned}$$

Solving (1) and (5) for p and q , we get

$$p = \sqrt{a^2 - x^2}, q = \frac{a^2}{y^2} - 1$$

Using these values of p and q in

$$dz = p dx + q dy,$$

we get

$$dz = \sqrt{a^2 - x^2} dx + \left[\frac{a^2}{y^2} - 1 \right] dy.$$

Integration implies

$$z = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) - \frac{a^2}{y} - y + b$$

Hence, the required complete integral is

$$z = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) - \frac{a^2}{y} - y + b$$

where a and b are arbitrary constants.

The Four Standard Forms

Standard Form I (Only p and q present)

In this form, we consider the equation of the form

$$f(p, q) = 0 \quad \dots \quad (1)$$

Charpit's Auxiliary equations are

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} \quad \dots \quad (2)$$

From (1), we have

$$f_x = 0, f_y = 0, f_z = 0, f_p \neq 0, f_q \neq 0$$

$\therefore$ using this in (2), we get

$$\frac{dp}{0} = \frac{dq}{0}$$

Taking first ratio, we get

$$dp = 0 \Rightarrow q = b \quad \dots \quad (4)$$

$\therefore$ (1) reduces to

$$f(a, b) = 0 \quad \dots \quad (5)$$

Using

$$dz = p dx + q dy$$

we get

$$dz = a dx + b dy$$

Integration implies

$$z = ax + by + C \quad \dots \quad (6)$$

where, C is an arbitrary constant.

Now, equations (5) and (6) together give the required solution.

Solving (5) for b , we get

$$b = F(a), \text{ say}$$

putting this value of b in (6) the required complete integral is

$$z = ax + yF(a) + C \quad \dots \quad (7)$$

To find singular solution of (1) we eliminate a and C between the complete integral (7) and the equations obtained by differentiating (7) partially w.r.t. a and C . That is, between the equations

$$z = ax + yF(a) + C, \quad 0 = x + yF'(a) \text{ and } 0 = 1 \quad \dots \quad (8)$$

Since the third relation of (8) is meaningless, we conclude that the PDE of standard form I has no singular solution.

To find the general solution of (1), we first take $c = \phi(a)$ in (7), where ϕ is an arbitrary function, and get.

$$z = ax + yF'(a) + \phi(a) \quad \dots \quad (10)$$

Eliminating a between (9) and (10) we get the required general solution.

Example

Solve $p^2 + q^2 = m^2$, where m is a constant.

Solution

The given equation is

$$p^2 + q^2 = m^2 \quad \dots \quad (1)$$

(1) is of the form

$$f(p, q) = 0$$

whose solution is

$$z = ax + by + c \quad \dots \quad (2)$$

where $a^2 + b^2 = m^2$

$$\therefore b = \sqrt{m^2 - a^2}$$

$\therefore$ the complete solution of (1) is

$$z = ax + y\sqrt{m^2 - a^2} + c \quad \dots \quad (3)$$

where a and c are arbitrary constants.

For Singular Solution

Differentiating (3) partially w.r.t. a and c , we get

$$0 = x - \frac{ay}{\sqrt{m^2 - a^2}} \text{ and } 0 = 1 \text{ but } 0 = 1$$

is meaning less. Hence, there is no singular solution of (1).

To Find the General Solution

Put $c = \phi(a)$ in (3). Then, we have

$$z = ax + y\sqrt{m^2 - a^2} + \phi(a) \quad \dots \quad (4)$$

Differentiating (4) w.r.t. a , we get

$$0 = x - \frac{ay}{\sqrt{m^2 - a^2}} + \phi'(a) \quad \dots \quad (5)$$

Eliminating a from (4) and (5) we get the required general solution of (1).

Standard Form II (Clairaut's Form)

A first order PDE is said to be in Clairaut's form if it can be written in the form

$$z = px + qy + f(p, q) \quad \dots \quad (1)$$

Let

$$F(x, y, z, p, q) = px + qy + f(p, q) - z = 0 \quad \dots \quad (2)$$

Charpit's A.E. are

$$\begin{aligned} \frac{dp}{F_x + pF_z} &= \frac{dq}{F_y + qF_z} = \frac{dz}{-pF_p - qF_q} = \frac{dx}{-F_p} = \frac{dy}{-F_q} \\ \Rightarrow \frac{dp}{0} &= \frac{dq}{0} = \frac{dz}{-px - qy - p\left(\frac{\partial F}{\partial p}\right) - q\left(\frac{\partial F}{\partial q}\right)} = \frac{dx}{-x - \frac{\partial F}{\partial p}} = \frac{dy}{-y - \frac{\partial F}{\partial q}} \end{aligned}$$

(Using (1))

Then, the first and second fractions imply

$$dp = 0 \text{ and } dq = 0$$

$$\Rightarrow p = a \text{ and } q = b.$$

Putting these values in (1) the required solution is

$$z = ax + by + f(a, b).$$

Examples

1. Solve $z = px + qy + c\sqrt{1 + p^2 + q^2}$

Solution

The given equation is

$$z = px + qy + c\sqrt{1 + p^2 + q^2} \quad \dots \quad (1)$$

which is of the form

$$z = px + qy + f(p, q) \quad \dots \quad (2)$$

Therefore, it is Clairaut's equation and so its complete integral is

$$z = ax + by + c\sqrt{1 + a^2 + b^2} \quad \dots \quad (3)$$

where a and b are arbitrary constants.

Singular Integral

Differentiating (3) partially w.r.t. a and b we get

$$\begin{aligned} 0 &= x + \frac{ac}{\sqrt{1 + a^2 + b^2}} \\ \Rightarrow x &= -\frac{ac}{\sqrt{1 + a^2 + b^2}} \quad \dots \quad (4) \end{aligned}$$

And

$$\begin{aligned} 0 &= y + \frac{bc}{\sqrt{1 + a^2 + b^2}} \\ \Rightarrow y &= -\frac{bc}{\sqrt{1 + a^2 + b^2}} \quad \dots \quad (5) \end{aligned}$$

From (4) and (5), squaring and adding, we get

$$\begin{aligned} x^2 + y^2 &= \frac{a^2c^2 + b^2c^2}{1 + a^2 + b^2} \\ \Rightarrow c^2 - x^2 - y^2 &= c^2 - \frac{a^2c^2 + b^2c^2}{1 + a^2 + b^2} \\ \Rightarrow c^2 - x^2 - y^2 &= \frac{c^2}{1 + a^2 + b^2} \quad \dots \quad (6) \end{aligned}$$

From (4)

$$a = -\frac{x\sqrt{1 + a^2 + b^2}}{c} = \frac{-x}{\sqrt{c^2 - x^2 - y^2}}, \quad \text{using (6)}$$

Similarly from (5) and (6), we get

$$b = \frac{-y}{\sqrt{c^2 - x^2 - y^2}}$$

Putting these values of a and b in (1), the required singular solution is

$$\begin{aligned} z &= \frac{-x^2}{\sqrt{c^2 - x^2 - y^2}} - \frac{y^2}{\sqrt{c^2 - x^2 - y^2}} + \frac{c^2}{\sqrt{c^2 - x^2 - y^2}} \\ &= \frac{c^2 - x^2 - y^2}{\sqrt{c^2 - x^2 - y^2}} \\ &= \sqrt{c^2 - x^2 - y^2} \\ \Rightarrow z^2 &= c^2 - x^2 - y^2 \\ \Rightarrow x^2 + y^2 + z^2 &= c^2 \end{aligned}$$

General Solution

Let $b = \phi(a)$, where ϕ is an arbitrary function. The equation (1) reduces to

$$z = ax + y\phi(a) + c\sqrt{1 + a^2 + [\phi(a)]^2} \quad \dots \quad (7)$$

Differentiating (7) partially w.r.t. a , we get

$$0 = x + y\phi'(a) + \frac{c[2a + 2\phi(a)\phi'(a)]}{2\sqrt{1 + a^2 + [\phi(a)]^2}} \quad \dots \quad (8)$$

Eliminating a between (7) and (8), the required general solution is obtained.

2. Find the complete and singular integral of

$$z = px + qy + \ln(pq)$$

Solution

The given equation is

$$z = px + qy + \ln(pq) \quad \dots \quad (1)$$

which is of the form

$$z = px + qy + f(p, q)$$

and so it is Clairaut's equation.

$\therefore$ Its complete integral is

$$\begin{aligned} z &= ax + by + \ln(ab) \\ \Rightarrow z &= ax + by + \ln a + \ln b \quad \dots \quad (2) \end{aligned}$$

where a and b are arbitrary constants.

Differentiating (2) partially w.r.t. a and b we get

$$0 = x + \frac{1}{a} \Rightarrow a = -\frac{1}{x} \quad \dots \quad (3)$$

And

$$0 = y + \frac{1}{b} \Rightarrow b = -\frac{1}{y} \quad \dots \quad (4)$$

Eliminating a and b from (2), (3) and (4) we get

$$z = -1 - 1 + \ln\left(\frac{1}{xy}\right) = -2 - \ln(xy)$$

$\therefore$ the required singular integral is

$$z = -2 - \ln(xy).$$

Stand form III (Only p, q and z Present]

In this form the differential equation is of the form

$$f(p, q, z) = 0 \quad \dots \quad (1)$$

Charpit's A.E. are

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

using (1), we get

$$\frac{dp}{pf_z} = \frac{dq}{qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

Taking first two ratios, we have

$$\begin{aligned} \frac{dp}{pf_z} &= \frac{dq}{qf_z} \\ \Rightarrow \frac{1}{p} dp &= \frac{1}{q} dq \end{aligned}$$

Integration implies

$$q = ap \quad \dots \quad (2)$$

where a is an arbitrary constant.

Now,

$$\begin{aligned}
dz &= p dx + a p dy \\
\Rightarrow dz &= p(dx + a dy) \\
\Rightarrow dz &= p d(x + ay) \\
\Rightarrow dz &= p du \quad \dots \quad (3)
\end{aligned}$$

where $u = x + ay \quad \dots \quad (4)$

Now, (3) implies

$$p = \frac{dz}{du}$$

and using (2), we get

$$q = ap = a \frac{dz}{du}$$

Substituting these values of p and q in (1), we get

$$f\left(\frac{dz}{du}, a \frac{dz}{du}, z\right) = 0 \quad \dots \quad (5)$$

(5) is now an ordinary differential equation of first order. Solving (5) we get z as a function of u and replacing u by $x + ay$, we get the required complete integral.

Example

1. Find the complete integral of $9(p^2 z + q^2) = 4$

Solution

The given equation is

$$9(p^2 z + q^2) = 4 \quad \dots \quad (1)$$

which is of the form

$$f(p, q, z) = 0$$

Let $u = x + ay$, where a is an arbitrary constant.

Now, putting $p = \frac{dz}{du}$ and $q = a \frac{dz}{du}$ in (1), we get

$$\begin{aligned}
9 \left[z \left(\frac{dz}{du} \right)^2 + a^2 \left(\frac{dz}{du} \right)^2 \right] &= 4 \\
\Rightarrow \left(\frac{dz}{du} \right)^2 &= \frac{4}{9(z + a^2)} \\
\Rightarrow \frac{du}{dz} &= \pm \frac{3}{2} \sqrt{z + a^2} \\
\Rightarrow du &= \pm \frac{3}{2} \sqrt{z + a^2} dz \\
\Rightarrow u + b &= \pm \frac{3}{2} \times \frac{(z + a^2)^{\frac{3}{2}}}{\frac{3}{2}} \\
\Rightarrow (u + b)^2 &= (z + a^2)^3 \\
\Rightarrow (x + ay + b)^2 &= (z + a^2)^3
\end{aligned}$$

which is the required complete integral.

Example

Find the complete integral of $p^2 + q^3 - 3pqz = 0$

Solutoin

The given equation is

$$p^2 + q^3 - 3pqz = 0 \quad \dots \quad (1)$$

which is of the form $f(p, q, z) = 0$

So, let $u = x + ay$, where a is an arbitrary constant now replacing p and q by $\frac{dz}{du}$ and $a \frac{dz}{du}$ in (1), we get

$$\left(\frac{dz}{du}\right)^3 + a^3 \left(\frac{dz}{du}\right)^3 - 3az \left(\frac{dz}{du}\right)^2 = 0$$

$$\text{or, } (1 + a^3) \frac{dz}{du} = 3az$$

$$\text{or, } \frac{1 + a^3}{z} dz = 3adu$$

Integration implies

$$(1 + a^3) \ln z = 3au + b$$

$$\Rightarrow (1 + a^3) \ln z = 3a(x + ay) + b,$$

which is the required complete integral.

Standard Form IV (Equation of the Form $f_1(x, p) = f_2(y, q)$)

Let

$$F(x, y, z, p, q) = f_1(x, p) - f_2(y, q) \quad \dots \quad (1)$$

Charpit's A.E. are

$$\frac{dp}{F_x + pF_z} = \frac{dq}{F_y + qF_z} = \frac{dz}{-pF_p - qF_q} = \frac{dx}{-F_p} = \frac{dy}{-F_q}$$

Using (1) we get

$$\frac{dp}{\frac{\partial f_1}{\partial x}} = \frac{dq}{-\frac{\partial f_2}{\partial x}} = \frac{dz}{-p \frac{\partial f_1}{\partial p} + q \frac{\partial f_2}{\partial q}} = \frac{dx}{-\frac{\partial f_1}{\partial p}} = \frac{dy}{\frac{\partial f_2}{\partial q}}$$

Taking first and fourth ratios, we get

$$\frac{\frac{dp}{\frac{\partial f_1}{\partial x}}}{-\frac{\partial f_1}{\partial p}} = \frac{\frac{dx}{-\frac{\partial f_1}{\partial p}}}{-\frac{\partial f_1}{\partial p}}$$

$$\Rightarrow \frac{\partial f_1}{\partial p} dp + \frac{\partial f_1}{\partial x} dx = 0$$

$$\Rightarrow df_1 = 0$$

Integration implies

$$f_1 = a$$

where, a is an arbitrary constant.

Therefore, (1) implies

$$f_1(x, p) = a = f_2(y, q) \quad \dots \quad (2)$$

That is,

$$f_1(x, p) = a \text{ and } f_2(y, q) \quad \dots \quad (3)$$

Solving (3) for p and q we get

$$p = F_1(x, a) \text{ and } q = F_2(y, a), \text{ say} \quad \dots \quad (4)$$

Substituting these values in

$$dz = p dx + q dy$$

we get

$$dz = F_1(x, a) dx + F_2(y, a) dy$$

Integration implies

$$z = \int F_1(x, a) dx + \int F_2(y, a) dy + b$$

where, a and b are arbitrary constants

$\therefore$ the complete integral is

$$z = \int F_1(x, a) dx + \int F_2(y, a) dy + b.$$

Example

Find the complete integral of $x(1+y)p = y(1+x)q$

Solution

The given equation is:

$$\begin{aligned} x(1+y)p &= y(1+x)q \\ \Rightarrow \frac{x}{1+x}p &= \frac{y}{1+y}q \quad \dots \quad (1) \end{aligned}$$

which is of the form

$$\begin{aligned} f_1(x, p) &= f_2(y, q) \\ \therefore f_1(x, p) &= a = f_2(y, q) \\ \Rightarrow \frac{x}{1+x}p &= a \quad \text{and} \quad \frac{y}{1+y}q = a \\ \Rightarrow p &= a \frac{1+x}{x} \quad \text{and} \quad q = a \frac{1+y}{y} \end{aligned}$$

Putting these values of p and q in

$$dz = p dx + q dy$$

we get

$$\begin{aligned} dz &= a \frac{1+x}{x} dx + a \frac{1+y}{y} dy \\ \Rightarrow dz &= a \left(\frac{1}{x} + 1 \right) dx + a \left(\frac{1}{y} + 1 \right) dy \end{aligned}$$

Integration implies

$$\begin{aligned} z &= a(\ln x + x) + a(\ln y + y) + b \\ \Rightarrow z &= a(\ln xy + x + y) + b \end{aligned}$$

which is the required complete integral where a and b are arbitrary constants.

Example

Solve $py + qx + pq = 0$

Solution

The given equation is

$$\begin{aligned} py + qx + pq &= 0 \\ \text{or, } py + q(x+p) &= 0 \\ \text{or, } \frac{p}{(x+p)} + \frac{q}{y} &= 0 \\ \text{or, } \frac{p}{(x+p)} &= -\frac{q}{y} \quad \dots \quad (1) \end{aligned}$$

which is of the form

$$\begin{aligned} f_1(x, p) &= f_2(y, q) \\ \therefore f_1(x, p) &= a = f_2(y, q) \\ \Rightarrow \frac{p}{(x+p)} &= a \quad \text{and} \quad -\frac{q}{y} = a \\ \Rightarrow p &= a(x+p) \quad \text{and} \quad q = -ay \\ \Rightarrow p &= \frac{ax}{1-a} \quad \text{and} \quad q = -ay \end{aligned}$$

Putting these values of p and q in

$$dz = p dx + q dy$$

we get

$$dz = \left(\frac{ax}{1-a} \right) dx - ay dy$$

Integration implies

$$z = \left(\frac{a}{1-a} \right) \frac{x^2}{2} - a \frac{y^2}{2} + \frac{b}{2}$$

$$2z = \left(\frac{a}{1-a} \right) x^2 - ay^2 + b$$

which is the required complete integral with two arbitrary constants a and b .